

# Supplementary Information for

## Recipe determinants of quantization fragility under image corruption:

### paired evidence and a regression-head localization in object detectors

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## S1 Purpose and scope

This supplement documents the frozen protocol details, estimands, corruption parameters, calibration schedules, and artifact contract supporting the main article. All reported values are generated deterministically from retained ledgers. Section S8 additionally reports per-family  $\Delta E$  diagnostics that support—but do not extend—the main-text claims.

## S2 Estimands

For dataset  $d$ , model  $m$ , corruption condition  $c$ , and a pair of treatments (A, B) sharing the same evaluator:

$$G_0 = \text{AP}_A(\text{J95}) - \text{AP}_B(\text{J95}), \quad (\text{S1})$$

$$G_c = \overline{\text{AP}_A(c) - \text{AP}_B(c)} \quad (\text{equal weight over the 12 cells}), \quad (\text{S2})$$

$$\Delta E = G_c - G_0. \quad (\text{S3})$$

J95 denotes the codec-matched clean control: the J95 clean control and every corrupted input are terminally encoded as JPEG quality 95 with chroma subsampling disabled, so the clean control and the corrupted inputs differ only in the named corruption transform, not in container encoding; the separate original-clean cell keeps the dataset-native bytes and is used only to check control fidelity.  $\Delta E > 0$  means the A-minus-B gap widens under corruption;  $\Delta E < 0$  means it narrows. A narrowing gap near a shared accuracy floor is not read as robustness: absolute corrupted AP is reported alongside every interaction (main text, §4.1–§4.3).

## S3 Frozen condition schedule

Each dataset×model block evaluates 14 conditions on identical image ordering for quantized arms (13 for FP32 and INT8-legacy arms that predate codec control):

- clean (original bytes) and J95 (JPEG-95 clean control);
- Gaussian noise:  $\sigma \in \{8, 20, 40\}$ ;
- motion blur: kernel lengths  $\{5, 13, 23\}$ ;
- synthetic fog: (strength, radius)  $\in \{(0.18, 6), (0.38, 12), (0.62, 20)\}$ ;
- JPEG compression: qualities  $\{70, 40, 15\}$ .

Corruption parameters are frozen in versioned configuration `nn.paired_protocol_v1.20260924`; the same transform code and the same JPEG-95 terminal encoding produce both the evaluation caches and the corruption-aware calibration images.

## S4 Treatments and arm registry

Table S1: RetinaNet arm registry. “Calibration” lists the input distribution seen by entropy calibration; “sites” lists the quantized region.

| Arm                   | Precision | Quantized sites                                | Calibration input                                 |
|-----------------------|-----------|------------------------------------------------|---------------------------------------------------|
| FP32                  | FP32      | none                                           | —                                                 |
| FP8-matched           | FP8       | full graph                                     | clean, matched preprocessing                      |
| INT8-legacy           | INT8      | full graph                                     | mismatched preprocessing (diagnostic)             |
| INT8-matched          | INT8      | full graph                                     | clean, 512 images                                 |
| INT8-q95calib         | INT8      | full graph                                     | J95 re-encodes, 512 images                        |
| INT8-selective        | INT8      | all except <code>/head/regression_head/</code> | clean, 512 images                                 |
| INT8-sel-q95calib     | INT8      | selective                                      | J95 re-encodes, 512 images                        |
| INT8-corruptcalib     | INT8      | full graph                                     | fold-1 corruptions (noise, JPEG), 512 images      |
| INT8-sel-corruptcalib | INT8      | selective                                      | in-family corruptions, 512 images                 |
| INT8-cc2calib         | INT8      | full graph                                     | fold-2 corruptions (fog, motion blur), 512 images |
| INT8-sel-cc2calib     | INT8      | selective                                      | fold-2 corruptions, 512 images                    |

All INT8/FP8 engines are built by TensorRT 11.1.0.106 from ONNX graphs quantized by NVIDIA ModelOpt 0.45.0 (entropy calibration, 512-image budget) with a 4,096 MiB builder workspace. Builder policy is recorded per engine and differs by program: the YOLO11 engines were built by `trtexec` before the frozen contract and carry TensorRT 11.1 defaults (TF32 enabled), while every RetinaNet arm built for this study passes `--noTF32`. Calibration and evaluation images are disjoint. Checkpoints are one per dataset×model cell, and their selection provenance is disclosed: the KITTI and VOC YOLO11n/m/x models were fine-tuned from Ultralytics pretrained weights under the pilot program (`*_train_v1` runs, dataset YAMLS with `val: images/val`), with the best checkpoint taken on the same partition that serves as the evaluation universe; COCO YOLO11 uses the official pretrained checkpoint without a dataset fine-tune; the torchvision RetinaNet-R50-FPN-v2 model is fine-tuned per dataset for 100 epochs at 640-pixel input (learning rate  $10^{-4}$ , weight decay  $10^{-4}$ , deterministic seed 20260807), selecting the best checkpoint by minimum validation loss on the same evaluation partition. Per-run training manifests are hash-bound; the evaluation-partition overlap is a disclosed selection caveat (Section ??). The frozen condition schedule is 14 conditions per arm; the RetinaNet FP32 and legacy arms predate codec control and carry 13.

## S5 Inference and evaluation contract

YOLO11 blocks: 640-pixel letterbox input, confidence threshold 0.001, class-wise NMS at IoU 0.7, at most 300 detections. RetinaNet blocks: 640-pixel letterbox input, confidence threshold 0.05, batched NMS at IoU 0.5, at most 300 detections. All evaluations use the COCO protocol via `pycocotools` with `maxDets=100`; VOC and KITTI annotations were converted once to COCO format and frozen. Decoder constants differ across families and are held fixed within each; cross-family comparisons are comparisons of recipe bundles, never of formats alone.

## S6 Corruption-aware calibration schedule

The intervention calibration sets reuse the same 512 images as the matched clean-calibration manifests. Each image is deterministically assigned—via a SHA-256-keyed hash of its identifier—to exactly one of six in-family cells, Gaussian noise or JPEG at severities 1, 3, or 5 (approximately 77–102 images per cell). Assigned corruptions use the frozen transform parameters above, and the outputs are terminally encoded at JPEG quality 95, matching the evaluation input contract. In this first fold fog and motion blur never appear in calibration and form the held-out split. A complementary second fold (schedule `nn-corruptcalib-fold2-v1`, frozen before any of its runs) assigns the same images to fog or motion blur at severities 1, 3, or 5 (75–95 images per cell) and holds out Gaussian noise and JPEG. Its pre-specified rules are (i) a held-out gain over INT8-matched with an interval excluding zero and (ii) a positive held-out  $\Delta E$  with an interval excluding zero, against both INT8-matched and INT8-q95calib.

The codec-matched control manifests encode the same 512 images at JPEG quality 95 with no named corruption, isolating codec adaptation from corruption coverage.

## S7 Paired bootstrap

Within each dataset×model block, one deterministic schedule of  $B = 2,000$  resamples of image indices is shared across all arms and all conditions, so every contrast is evaluated inside common-image samples. Reported point estimates are full-sample plug-in values; reported intervals are the 2.5th–97.5th percentiles of the bootstrap

distribution of each contrast. Intervals cover image resampling only; they do not cover training-seed, calibration-draw, engine-build, or corruption-realization variance, and image-level resampling treats frames as independent (KITTI frames from shared driving sequences are correlated, so those intervals may be optimistic).

Two-sided bootstrap  $p$ -values are  $p = \min\{1, 2 \min[\Pr(d \leq 0), \Pr(d \geq 0)]\}$  (two-sided resolution 1/1000) and are Holm-adjusted across the nine paired YOLO11 blocks. Heterogeneity of  $\Delta E$  across those blocks uses Cochran’s  $Q = \sum_i w_i (\hat{\Delta}_i - \bar{\Delta}_w)^2$  with  $w_i = 1/\hat{\text{se}}_i^2$  from the bootstrap draws,  $I^2 = \max\{0, (Q - \text{df})/Q\}$ , and the DerSimonian–Laird between-block standard deviation  $\tau$ . Blocks within a dataset share images, so their estimates are plausibly positively correlated (the cross-block covariance is not estimated); treating them as independent inflates the variance of between-block differences and makes  $Q$  and the pairwise  $z$  statistics conservative. RetinaNet and intervention intervals are per contrast.

Point estimates are full-sample plug-in values computed from the metric records of each cell (for COCO, from the frozen 2,000-image subset evaluation); bootstrap draws supply intervals, standard errors, and  $p$ -values only. For the intervention, the corruption-specific component on the held-out family is  $\Delta E_{\text{held-out}} = (\text{AP}_A - \text{AP}_B)_{\text{held-out}} - (\text{AP}_A - \text{AP}_B)_{\text{J95}}$ , the same paired construction as the main estimand. This decomposition is interpretive: it assumes an additive, parallel-shift model in which a calibration-induced level shift on clean inputs transfers comparably to corrupted inputs. Under an interaction model (where the intervention also changes slope across conditions),  $\Delta E_{\text{held-out}}$  mixes that interaction with any clean-level effect; it is not a causal identification theorem. We therefore report the J95-clean and held-out levels alongside it rather than relying on the composite alone.

## S8 Sensitivity analyses

Table S2 resolves the paired  $\Delta E$  of each YOLO11 block per corruption family (each entry is the family-mean FP8–INT8 gap minus the J95 gap) and gives the leave-one-out exclusion of the `motion_blur-s5` severity wall. The sign heterogeneity is also *within* blocks: on VOC-x the JPEG family contributes  $-1.07$  AP while Gaussian noise and motion blur contribute  $+0.7$  AP each; on COCO-m JPEG and motion blur contribute positively while noise contributes a null negative. Motion blur is the most block-variable family of the four (across-block standard deviation  $\approx 1.1$  AP population,  $\approx 1.2$  sample), with JPEG close behind ( $\approx 1.0$ ); neither is uniformly harmful or helpful. Excluding the severity wall leaves the cross-block heterogeneity essentially unchanged ( $Q \approx 60.2$ ,  $I^2 \approx 87\%$ ) and every Holm-surviving block’s sign intact; the two near-null KITTI blocks drift (YOLO11n flips from  $-0.01$  to  $+0.15$ , still null, and YOLO11m’s negative effect loses significance). The interaction pattern is therefore not an artifact of the saturated cell, although the marginal KITTI estimates are sensitive to it.

The same leave-one-out sensitivity applied to the intervention contrasts changes no frozen-criterion verdict; two secondary codec-control contrasts do move across the zero boundary on VOC (both on the held-out basis): INT8-corruptcalib minus INT8-q95calib goes from  $+0.13$   $[-0.02, +0.28]$  to  $+0.21$   $[+0.04, +0.39]$ , and INT8-sel-q95calib minus INT8-selective goes from  $-0.11$   $[-0.23, +0.01]$  to  $-0.15$   $[-0.29, -0.01]$ . Both are read with the main text’s no-multiplicity caveat.

The frozen configuration files are hash-identified: `nn_corruptcalib_v1_20260924.json` SHA-256 `f51011ca5d666b89...` and `nn_corruptcalib_fold2_v1_20260925.json` SHA-256 `5f2d122ae0a52d5f...` (full hashes in the archive manifest). Freeze timing is established by local-file evidence rather than an external registry: the versioned configuration files’ write times, the per-run manifest `created_at` fields, and the repository commit ordering all place the configurations before their arms’ first inference. The fold-1 file was written 2026-09-25 (its name retains the design date 20260924); the q95 codec-control schedules were added after the fold-1 results were computed—the attempt identifier `nn_q95calib_v1_20260924` keeps the design-date convention and postdates the fold-1 runs in manifest time. The fold-2 primary rule is quoted verbatim: “`int8-cc2calib512 minus int8-matched512 on the fold-2 held-out mean (gaussian_noise+jpeg), paired bootstrap B=2000, same seed namespace as Phase B: CI95 excluding 0 => fold-1 frozen criterion replicated on fold 2`”; we read “replicated” through the directional fold-1 criterion it cites, so a significantly negative change counts as failure.

Table S2: Per-family  $\Delta E$  (AP points; family-mean FP8–INT8 gap minus J95 gap) and the leave-one-out  $\Delta E$  excluding `motion_blur-s5`, per dataset×model block. \* interval excludes zero.

| Dataset | Model   | Fog       | Gaussian noise | JPEG      | Motion blur | $\Delta E$ excl. mb-s5     |
|---------|---------|-----------|----------------|-----------|-------------|----------------------------|
| KITTI   | YOLO11n | $+1.11^*$ | $+0.41$        | $+0.17$   | $-1.71^*$   | $+0.15$ $[-0.51, +1.03]$   |
| KITTI   | YOLO11m | $+0.38$   | $-0.44$        | $+0.02$   | $-2.68^*$   | $-0.35$ $[-0.98, +0.29]$   |
| KITTI   | YOLO11x | $-0.40$   | $-1.98^*$      | $-0.43$   | $+0.03$     | $-0.65^*$ $[-1.08, -0.29]$ |
| VOC     | YOLO11n | $+0.35^*$ | $+0.28$        | $+0.27$   | $+0.34^*$   | $+0.34^*$ $[+0.09, +0.62]$ |
| VOC     | YOLO11m | $-0.03$   | $-0.48^*$      | $-1.23^*$ | $-0.56^*$   | $-0.53^*$ $[-0.76, -0.31]$ |
| VOC     | YOLO11x | $+0.26^*$ | $+0.69^*$      | $-1.07^*$ | $+0.67^*$   | $+0.13$ $[-0.06, +0.35]$   |
| COCO-2k | YOLO11n | $+0.17$   | $-0.22$        | $-0.56^*$ | $-0.69^*$   | $-0.28$ $[-0.63, +0.05]$   |
| COCO-2k | YOLO11m | $+0.36^*$ | $-0.36$        | $+1.43^*$ | $+0.99^*$   | $+0.60^*$ $[+0.27, +0.97]$ |
| COCO-2k | YOLO11x | $+0.15$   | $-0.28$        | $-2.10^*$ | $+0.13$     | $-0.59^*$ $[-1.00, -0.26]$ |

## S9 Evidence layers

Diagnostic arms (the RetinaNet decomposition: head-only and all-except-region variants) are single-point recipe sweeps produced during development and carry no intervals. Frozen paired evidence is the 14-condition paired protocol; its freeze covers the estimands, condition schedule, bootstrap schedule, and multiplicity adjustment, while the YOLO11 engines and evaluation partitions predate it and are shared with the historical layer. Intervention evidence is the corrupt-calibration experiment together with its codec-matched controls, whose folds were frozen prospectively before their engines were built. Historical exploratory results (a 144-cell grid over four datasets including TT100K, computed on the same engines under a different estimand and partitions) are reported only as context and are never pooled with the frozen paired estimates.

## S10 Artifact and audit boundary

Manuscript tables and figures are generated deterministically from frozen ledgers by `analysis/build_nn_tables.py`. Engine registries, calibration manifests (including the corrupt-calibration cell-assignment schedule), input manifests, prediction manifests, metric records, and paired-bootstrap draw caches are SHA-256 hash-bound and retained in the versioned research archive (concept DOI <https://doi.org/10.5281/zenodo.22031663>). Raw benchmark datasets, trained checkpoints, built TensorRT engines, and raw predictions are not redistributed; their provenance and regeneration path is documented in the archive.

## S11 Minimum reporting checklist for paired corruption studies

- record the complete treatment recipe (checkpoint, quantizer coverage, calibration list and preprocessing, builder settings, decoder), not just the datatype label;
- pair treatments on identical image bytes, including terminal encoding;
- subtract the matched clean gap before reading a corruption interaction;
- report absolute corrupted AP and the clean gap alongside every interaction, so floor effects cannot masquerade as robustness;
- keep calibration corruption families disjoint from held-out evaluation families when claiming calibration-based robustness;
- carry codec-matched calibration controls before attributing calibration-intervention gains to corruption coverage;
- flag severity-wall cells where every treatment including FP32 is at the floor, and report a leave-one-out sensitivity.